

CloudShift - Software Management | Project Plan Report

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1. Planning Basis

The case supplies 9 application effort figures and a 3-engineer team. The PERT workbook gives a theoretical migration duration of 42 working days from 126 person-days / 3 engineers. That figure is a productive-effort estimate, not a complete calendar schedule.

For scheduling, the three migration waves are planned with one engineer primarily assigned to each application in a wave. Therefore, the wave duration is governed by the longest application effort in that wave:

- Wave 1 (Apps 1-3): $\max(8, 10, 6) = 10$ working days
- Wave 2 (Apps 4-6): $\max(12, 9, 15) = 15$ working days
- Wave 3 (Apps 7-9): $\max(22, 22, 22) = 22$ working days, using the PERT expected effort of 22 person-days for each undocumented app

This creates a transparent distinction between 42 working days of theoretical balanced effort and the resource-constrained wave schedule.

2. Work Breakdown Structure (WBS)

None

1.0 CloudShift Application Migration Project

- └─ 1.1 Planning, Discovery & Environment Setup
 - | └─ 1.1.1 Requirements elicitation & sign-off
 - | └─ 1.1.2 Legacy infrastructure/application audit
 - | └─ 1.1.3 Cloud landing zone & IAM foundation
- └─ 1.2 IaC & Data Pipeline Foundation
 - | └─ 1.2.1 Reusable IaC modules
 - | └─ 1.2.2 CDC replication pipeline
 - | └─ 1.2.3 Blackout Guard
- └─ 1.3 Wave-Based Migration

- | └─ 1.3.1 Wave 1: Apps 1-3
- | └─ 1.3.2 Wave 2: Apps 4-6
- | └─ 1.3.3 Wave 3: Apps 7-9
- └─ 1.4 Verification & Cutover
 - | └─ 1.4.1 SHA-256 validation
 - | └─ 1.4.2 Blackout verification & DNS cutover
 - | └─ 1.4.3 Synthetic checks & rollback
- └─ 1.5 Handover & Decommissioning
 - └─ 1.5.1 Operational documentation/sign-off
 - └─ 1.5.2 Legacy server decommissioning

3. Resource Allocation Matrix

Engineer	Primary Role	Wave 1	Wave 2	Wave 3
Eng 1	DB Specialist	App2 - 10d	App4 - 12d	App9 - 22d
Eng 2	Cloud Infrastructure	App1 - 8d	App6 - 15d	App7 - 22d
Eng 3	QA/Application Lead	App3 - 6d	App5 - 9d	App8 - 22d

The assignments use the supplied per-application figures. Wave duration is the maximum parallel effort within the wave, not the sum of the three engineer assignments.

4. Critical Path Method (CPM) & Network Data

4.1 Task Network

Task	Description	Predecessor	Duration	Critical?
T01	Discovery & requirements sign-off	None	3d	Yes
T02	Cloud landing zone & IaC foundation	T01	3d	Yes
T03	Wave 1 migration (Apps 1-3)	T02	10d	Yes
T04	Wave 1 cutover & verification	T03	2d	Yes
T05	Wave 2 migration (Apps 4-6)	T04	15d	Yes
T06	Wave 2 cutover & verification	T05	2d	Yes

T07	Wave 3 migration (Apps 7-9)	T06	22d	Yes
T08	Wave 3 cutover & verification	T07	3d	Yes
T09	Decommissioning & project handover	T08	2d	Yes
T10	Non-blocking security audit	T02	5d	No

The critical path is:

T01 -> T02 -> T03 -> T04 -> T05 -> T06 -> T07 -> T08 -> T09

Critical-path active duration:

$3 + 3 + 10 + 2 + 15 + 2 + 22 + 3 + 2 = 62$ working days.

T10 is parallel and is therefore non-critical as long as it is completed before the dependent security checkpoint.

5. CPM Forward/Backward Pass Analysis

Task	D	ES	EF	LS	LF	TF
T01	3	0	3	0	3	0
T02	3	3	6	3	6	0
T03	10	6	16	6	16	0
T04	2	16	18	16	18	0
T05	15	18	33	18	33	0
T06	2	33	35	33	35	0
T07	22	35	57	35	57	0
T08	3	57	60	57	60	0
T09	2	60	62	60	62	0
T10	5	6	11	30	35	24

For the linear critical chain, all tasks have zero total float. T10 has 24 working days of total float relative to its required completion before T07.

6. Blackout-Adjusted Calendar Schedule

The case defines 22 working days per month, with the last 5 working days unavailable for migration/cutover. Therefore:

Effective migration capacity = $22 - 5 = 17$ working days/month.

To keep the schedule consistent with the case-study blackout rule, tasks are split around blackout windows where necessary. The two blackout periods shown below are calendar constraints, not additional engineering effort.

None

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title CloudShift Calendar Schedule

dateFormat X

axisFormat WD %s

section Month 1

T01 Discovery                :t01, 1, 3d
T02 IaC Foundation           :t02, after t01, 3d
T03 Wave 1 Migration         :t03, after t02, 10d
T04 Wave 1 Cutover           :t04, after t03, 2d
Blackout WD 18-22            :crit, bw1, 18, 5d

section Month 2

T05 Wave 2 Migration Part 1  :t05a, 23, 12d
T05 Wave 2 Migration Part 2  :t05b, 35, 3d
T06 Wave 2 Cutover           :t06, 38, 2d
T07 Wave 3 Migration Part 1  :t07a, 40, 0d
Blackout WD 40-44            :crit, bw2, 40, 5d

section Month 3

T07 Wave 3 Migration Part 2  :t07b, 45, 22d
T08 Wave 3 Cutover           :t08, 67, 3d
T09 Handover                 :t09, 70, 2d

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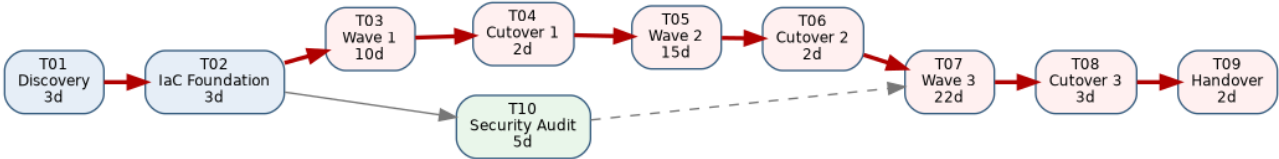
Schedule interpretation: the exact Gantt rendering should be treated as a sequencing illustration. The governing numerical result is the resource-constrained 62-day active schedule plus the month-end blackout constraints. The project remains well inside the 7-month hardware-support deadline.

7. Milestones

Milestone	Description	Checkpoint
MS-1	Cloud landing zone & IaC foundation ready	T02 complete
MS-2	Wave 1 production cutover complete	Apps 1-3 verified
MS-3	Wave 2 production cutover complete	Apps 4-6 verified
MS-4	Wave 3 production cutover complete	Apps 7-9 verified
MS-5	Project handover/decommission complete	T09 complete

All migration/cutover milestones must be scheduled outside the last 5 working days of a month.

Network Diagram and Critical Path



Gantt Chart

